Genetic heterogeneity of MCI/AD and its manifestation in the general population

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State-of-the-art data-driven machine learning methods have dissected neuroanatomical heterogeneity in Alzheimer's disease (AD) and its preclinical stage of mild cognitive impairment (MCI) (Vogel et al., 2021; Wen et al., 2021; Yang et al., 2021; Young et al., 2018; Zhang et al., 2016). However, the extent to which this neuroanatomical heterogeneity is attributable to underlying genetic heterogeneity remains unclear. We previously proposed a novel deep learning semi-supervised clustering method, termed Smile-GAN. Applying Smile-GAN to ADNI derived four subtypes in AD and MCI: P1 individuals with normal brain anatomy, P2 individuals with medial temporal-spared mild diffuse atrophy, P3 individuals with focal medial temporal lobe atrophy, and P4 individuals with advanced atrophy (Yang et al., 2021). The current study sought to elucidate the neuroanatomical manifestation and genetic profiles of the four MCI/AD signatures in the general population.

• Methods

T1-weighted images and imputed genotyping data were analyzed for individuals with Caucasian ancestry in UKBIOBANK [N = 17990 (age = 55.17±7.44 years; 52,32% female)]. For image processing, 145 regions of interest (ROIs) were extracted using a multi-atlas segmentation technique (Doshi et al., 2016). For genetic quality checks, we removed related (2nd-degree related individuals), sex-ambiguous, and duplicated individuals. Lastly, we excluded i) individuals with more than 3% missing genotypes; ii) variants with minor allele frequency less than 0.01; iii) variants with larger than 3% missing genotyping rate; iv) variants that failed the Hardy-Weinberge test at 10-10.

The Smile-GAN model trained on ADNI (Yang et al., 2021) was subsequently applied to UKBB ROIs to derive the expression scores (i.e., pseudo probability scores) of the four subtypes. We performed GWAS with multiple linear regressions controlling for age, sex, ICV, and the first 40 genetic PCs via Plink (Purcell et al., 2007) for the four expression scores. FUMA (Watanabe et al., 2017) was then used to annotate the independent significant variants and genomic risk loci. MAGMA (Leeuw et al., 2015) was used to perform gene analysis with GWAS summary statistics. We manually queried each genomic risk locus and significant gene on GWAS Catalog (Buniello et al., 2019) to determine if it's de novo or previously associated with any clinical trait.

• Results

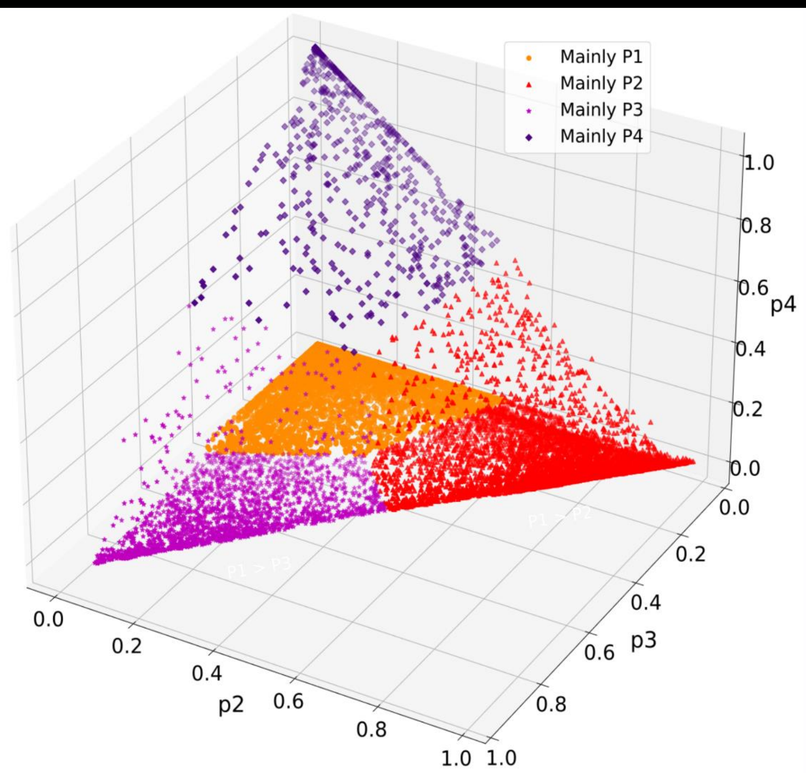

Figure 1. **Four AD subtypes manifest in the general population of UKBB**. Smile-GAN trained on ADNI was applied to UKBIOBANK, resulting in P1=11636 (orange circles), P2=4180 (red triangles), p3=1698 (magenta stars) and P4=476 (purple rectangles). The 3D scatter plot shows the pseudo probability (i.e, expression scores) of each individual belonging to each subtype. We define P1 individuals near the origin, which is analogous to the P1 individuals with normal brain anatomy in (Yang et al., 2021).

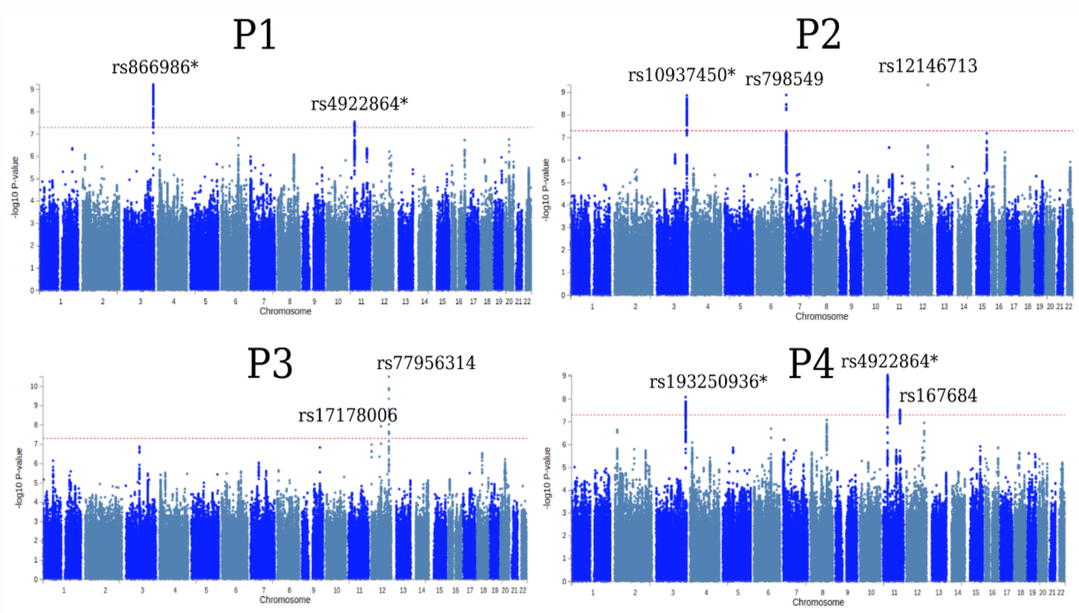

Figure 2. **The genetic architecture of the four AD subtypes in the general population**. We ran GWAS for the pseudo-probability scores of the four subtypes in UKBB and found 4 *de novo* (*) genomic loci. The genetic profile of the 4 subtypes shows considerable similarity and discrepancy. For instance, P1 and P4 detected similar genomic loci, but the effects are in the opposite direction (negatively correlated for their probability scores). Most independent significant variants are widely identified to be associated with brain volumes on GWAS Catalog.

Smile-GAN identified 11,636 P1 individuals, 4180 P2 individuals, 1698 P3 individuals, and 476 P4 individuals in the general population (Fig. 1). GWAS discovered ten genomic loci (9 unique), which passed the genome-wide significant threshold (-log10P-value > 7.3) for the four AD subtypes. Four genomic loci (rs866986, rs4922864, rs10937450, and rs193250936) were de novo on GWAS Catalog. P1 and P4 detected similar genetic loci but opposite effect directionality (i.e., beta values with opposite signs) (Fig. 2). Further correlation analysis revealed that their expression scores were negatively correlated (Spearman r = -0.95, P-value < 10-5). Gene analysis discovered significant associations for gene GMNC, DCDC1, DNAJ24 in P1; GMNC, AMZ1, GNA12, DCDC1 in P2, GMNC, DCDC1, DNAGC24 (de novo), TMEM135 in P4. GMNC and GNA12 were previously associated with P-tau and T-tau (Deming et al., 2017) and mood and psychotic disorders (Blokland et al., 2021), respectively.

• Conclusion

The current study evaluates the four neuroanatomical MCI/AD subtypes previously defined via a deep learning method in an independent sample. Notably, our results shed light on the genetic heterogeneity of these subtypes in the general population, indicating that partially distinct genetic risk factors might lead to heterogeneous clinical manifestations and underlying etiologies, and genetic heterogeneity might be crucial in predisposing an individual to AD.

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